

Abdullah Al Farhad

Natural Language Processing Researcher & Lecturer

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Research Profile

Natural Language Processing researcher and university lecturer with research experience in **chart summarization**, **logical reasoning** and **natural language inference**, **question answering/generation**, and multilingual language processing. Co-author of *ChartSumm*, a benchmark for automatic chart summarization. Experienced with transformer-based models including BERT, mBERT, T5, and SBERT, as well as data analysis, benchmarking, and model fine-tuning for NLP tasks. Interested in developing reliable language systems that combine structured reasoning, multimodal understanding, and faithful text generation.

Research Interests

Natural Language Processing • Chart-to-Text Generation and Chart Summarization • Natural Language Inference and Logical Reasoning • Question Answering and Question Generation • Multilingual NLP • Transformer-based Language Models • Multimodal NLP • Evaluation and Benchmarking

Education

Islamic University of Technology (IUT), Gazipur, Bangladesh

Jan 2018 – May 2022

B.Sc. in Computer Science and Engineering

CGPA: 3.76/4.00, First Class with Honors

- **Undergraduate Thesis:** *ChartSumm: A Large Benchmark for Chart Summarization*
- **Supervisor:** Dr. Abu Raihan Mostofa Kamal

Research Publication

R. Rahman, R. Hasan, A. Al Farhad, M. T. R. Laskar, M. H. Ashmafee, and A. R. M. Kamal.

“ChartSumm: A Comprehensive Benchmark for Automatic Chart Summarization of Long and Short Summaries.” *36th Canadian Conference on Artificial Intelligence*, 2023.

Research & Professional Experience

Green University of Bangladesh (GUB)

Feb 2023 – Present

Lecturer, Department of Computer Science and Engineering

Dhaka, Bangladesh

- Teach undergraduate courses and laboratories across Natural Language Processing, Machine Learning, Data Mining, Algorithms, Programming, Operating Systems, Web Programming, Data Communication, and related CSE areas.
- Supervise and support student projects involving machine learning, NLP, software systems, and applied data analysis.
- Contribute to academic activities, curriculum delivery, assessment, and departmental service.

Pioneer Alpha Ltd.

Aug 2021 – Jan 2022

Machine Learning Engineer Intern

Dhaka, Bangladesh

- Worked with the Question Generation team on transformer-based NLP models including BERT, mBERT, T5, and related architectures.
- Fine-tuned T5 for question-answering tasks and investigated methods for **logical reasoning question answering**.
- Conducted model and dataset exploration for question generation and reasoning-oriented NLP workflows.

Pioneer Alpha Ltd.

May 2021 – Jul 2021

Data Analyst Intern

Dhaka, Bangladesh

- Explored SQuAD 1.0/1.1, TyDi QA, and Bangla SQuAD datasets for NLP experimentation.
- Performed exploratory data analysis and topic/keyword modeling using SBERT, LDA, YAKE, and RAKE for the Question Generation team.

Selected Research & Technical Projects

- **ChartSumm – Chart Summarization Benchmark:** Benchmark development for automatic generation of long and short natural-language summaries from charts; undergraduate thesis and peer-reviewed conference publication.
- **Logical Reasoning Question Answering:** Investigated transformer-based approaches for reasoning-focused question answering during an ML engineering internship.
- **Multimodal Rumour Scanner** (Python, JavaScript), 2023–2024: multimodal system project for analysing potentially misleading or rumour-related content.
- **Aggregator and Analyser of Reviews** (Python, JavaScript): review aggregation and text-analysis application.
- **IUTPC Rating Tracker** (Python): web-scraping based rating tracker.
- **Stock Risk Analyzer** (Java), 2024–2025: software project for risk-oriented stock analysis.

Technical Skills

Programming: Python, C, C++, Java, C#, SQL, PL/SQL, JavaScript, PHP, MATLAB, Unix Shell Scripting, Assembly

NLP / ML: BERT, mBERT, T5, SBERT, transformer-based modeling, question answering, question generation, topic modeling, keyword extraction, exploratory data analysis

Web / Software: HTML, CSS, JavaScript, PHP, Unity, Jira Software, NS-2

Tools: MATLAB, AutoCAD, Blender, Adobe Illustrator, Adobe Photoshop, CyberLink PowerDirector

Selected Teaching Portfolio

NLP / AI: CSE 413 Natural Language Processing; CSE 414 Natural Language Processing Lab; CSE 412 Machine Learning Lab; CSE 435 Data Mining; CSE 436 Data Mining Lab.

Core CSE: Discrete Mathematics; Structured Programming; Object-Oriented Programming; Data Structures; Algorithms; Operating Systems; Computer Architecture; Digital Logic Design; Microprocessors and Microcontrollers; Compiler Lab.

Systems / Applications: Data Communication; Web Programming; Mobile Application Development; Integrated Design Project.

Training, Service & Activities

- Certificate Course in Teaching and Learning (CTL), Batch 25, Feb 2023 – Aug 2023.
- Organizer, ICPC Regional 2023; organizer, STI 2024 and 2025.
- Organizer, IUT National ICT Fest, 2018–2021; participant, Dhaka AI Challenge 2020.
- Completed Coursera coursework in Deep Learning, Machine Learning, and Data Science.
- Participated in Intra-IUT Programming Contests (2018, 2019, 2020); represented the CSE Department Cricket Team (2018–2021).

Languages

Bangla: Native

English: Fluent working proficiency (TOEFL: 84+)

References

Dr. Abu Raihan Mostofa Kamal

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